

# scGPT Human Validation — Reviewer 2 (Blinded)

## Purpose

Independently determine how each citing paper uses scGPT. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

## Workflow

Calibration: complete CAL-01 to CAL-03 independently, then discuss those three cases and freeze the codebook.

Formal validation: complete every HV row in the displayed order. Do not open the other reviewer document or the consensus document until both formal reviews are locked.

Reconciliation: after both reviewers lock their answers, the coordinator copies both records into the consensus document. Discuss only disagreements. Use `REFERENCE_UNRESOLVED` when agreement is not possible.

Evidence boundary: use only the linked approved article versions. Search all occurrences of “scGPT” and inspect the surrounding methods, results, figures, tables, and supplements that are linked.

## Allowed values

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located

Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

# Review table

| Article and approved evidence links                                                                                                                                                                                                                                                                         | Reviewer 2 independent record                                                                                                                                                                           |
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| <p>CAL-01   CALIBRATION<br/><b>scMGCL: accurate and efficient integration representation of single-cell multi-omics data</b><br/>2025   Bioinformatics<br/>DOI: 10.1093/bioinformatics/btaf392<br/><a href="#">Article page</a><br/><a href="#">Full text 1</a></p>                                         | <p>Review status: IN PROGRESS / LOCKED<br/>Citation validity:<br/>Method executed:<br/>Primary use:<br/>Biological insight:<br/>Evidence sufficiency:<br/>Supporting quote + locator:</p> <p>Notes:</p> |
| <p>CAL-02   CALIBRATION<br/><b>Genome language modeling (GLM): a beginner's cheat sheet</b><br/>2025   Biology Methods and Protocols<br/>DOI: 10.1093/biomethods/bpaf022<br/><a href="#">Article page</a><br/><a href="#">Full text 1</a></p>                                                               | <p>Review status: IN PROGRESS / LOCKED<br/>Citation validity:<br/>Method executed:<br/>Primary use:<br/>Biological insight:<br/>Evidence sufficiency:<br/>Supporting quote + locator:</p> <p>Notes:</p> |
| <p>CAL-03   CALIBRATION<br/><b>Building a learnable universal coordinate system for single-cell atlas with a joint-VAE model</b><br/>2024   Communications Biology<br/>DOI: 10.1038/s42003-024-06564-0<br/><a href="#">Article page</a><br/><a href="#">Full text 1</a><br/><a href="#">Full text 2</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/>Citation validity:<br/>Method executed:<br/>Primary use:<br/>Biological insight:<br/>Evidence sufficiency:<br/>Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-022   FORMAL VALIDATION<br/> <b>PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.02.26.707870<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                         | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-028   FORMAL VALIDATION<br/> <b>Single-cell sequencing in precision management of chronic myeloid leukemia</b><br/> 2026   Journal of Translational Genetics and Genomics<br/> DOI: 10.20517/jtgg.2026.42<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                            | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-012   FORMAL VALIDATION<br/> <b>MOSAIC: Intra-tumoral heterogeneity characterization through large-scale spatial and cell-resolved multi-omics profiling</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.05.15.654189<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-048   FORMAL VALIDATION<br/> <b>scBaseCount: an AI agent-curated, uniformly processed, and autonomously updated single cell data repository</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.02.27.640494<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-005   FORMAL VALIDATION<br/> <b>A multimodal cell-free RNA language model for liquid biopsy applications</b><br/> 2025   Nature Machine Intelligence<br/> DOI: 10.1038/s42256-025-01148-x<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                               | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-002   FORMAL VALIDATION<br/> <b>PertDiffBench: Benchmarking Diffusion Models for Single-Cell Perturbation Response Prediction</b><br/> 2026<br/> DOI: 10.64898/2026.06.13.732013<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                        | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-035   FORMAL VALIDATION<br/> <b>Revealing the Best Strategies for Rare Cell Type Detection in Multi-Sample Single-Cell Datasets</b><br/> 2025   Genes<br/> DOI: 10.3390/genes17010031<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                         | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-011   FORMAL VALIDATION<br/> <b>NanoCellAnnotator: Formalizing Expert Cell Type Annotation with Large Language Models</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.06.21.728965<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>            | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-045   FORMAL VALIDATION<br/> <b>Opportunities for artificial intelligence and synthetic biology in designing living drug delivery systems</b><br/> 2026   Advanced Drug Delivery Reviews<br/> DOI: 10.1016/j.addr.2026.115883<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-020   FORMAL VALIDATION<br/> <b>Challenges in AI-driven Biomedical Multimodal Data Fusion and Analysis</b><br/> 2025   Genomics Proteomics &amp; Bioinformatics<br/> DOI: 10.1093/gpbjnl/qzaf011<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                       | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-023   FORMAL VALIDATION<br/> <b>Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.11.13.688060<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a><br/> <a href="#">Full text 2</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-039   FORMAL VALIDATION<br/> <b>scGNN+: Adapting ChatGPT for Seamless Tutorial and Code Optimization</b><br/> 2024   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2024.09.30.615735<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                       | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-034   FORMAL VALIDATION<br/> <b>GET: a foundation model of transcription across human cell types</b><br/> 2023   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2023.09.24.559168<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-031   FORMAL VALIDATION<br/> <b>C5aR1 inhibition reprograms tumor associated macrophages and reverses PARP inhibitor resistance in breast cancer</b><br/> 2024   Nature Communications<br/> DOI: 10.1038/s41467-024-48637-y<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-040   FORMAL VALIDATION<br/> <b>Clifti-GPT: Privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data</b><br/> 2025<br/> DOI: 10.21203/rs.3.rs-7917089/v1<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>       | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-001   FORMAL VALIDATION<br/> <b>Generative genomics accurately predicts future experimental results</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.09.08.674753<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                     | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-038   FORMAL VALIDATION<br/> <b>cspray: Distributed Single Cell Transcriptome Analysis</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.02.06.704110<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                 | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-021   FORMAL VALIDATION<br/> <b>scaLR: a low-resource deep neural network-based platform for single cell analysis and biomarker discovery</b><br/> 2025   Briefings in Bioinformatics<br/> DOI: 10.1093/bib/bbaf243<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-047   FORMAL VALIDATION<br/> <b>A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches</b><br/> 2025   Briefings in Bioinformatics<br/> DOI: 10.1093/bib/bbaf355<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-015   FORMAL VALIDATION<br/> <b>scGPT-spatial: Continual Pretraining of Single-Cell Foundation Model for Spatial Transcriptomics</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.02.05.636714<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>   | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-017   FORMAL VALIDATION<br/> <b>CellMemory: hierarchical interpretation of out-of-distribution cells using bottlenecked transformer</b><br/> 2025   Genome biology<br/> DOI: 10.1186/s13059-025-03638-y<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                        | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-029   FORMAL VALIDATION<br/> <b>Elucidating the Design Space of Generative Models for Single-Cell Perturbation Prediction</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.06.15.732063<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                       | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-004   FORMAL VALIDATION<br/> <b>Discovering Candidate Anti-Aging Perturbations Using a Foundation Model for Gene Expression</b><br/> 2025   International Journal of Molecular Sciences<br/> DOI: 10.3390/ijms262411977<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                      | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-013   FORMAL VALIDATION<br/> <b>Integrating In silico perturbation with multilayer omics to decode regulatory networks in cancer immunity: a new frontier in precision oncology</b><br/> 2026   Frontiers in Immunology<br/> DOI: 10.3389/fimmu.2026.1857447<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-007   FORMAL VALIDATION<br/> <b>Benchmarking gene expression reconstruction from single-cell latent representations</b><br/> 2026<br/> DOI: 10.64898/2026.06.15.731445<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                    | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-025   FORMAL VALIDATION<br/> <b>Hybrid Attention-Enhanced Regularized Convolutional Autoencoder for Robust EEG Feature Extraction in Seizure Prediction</b><br/> 2025   IEEE Access<br/> DOI: 10.1109/access.2025.3608302<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-041   FORMAL VALIDATION<br/> <b>Nicheformer: a foundation model for single-cell and spatial omics</b><br/> 2025   Nature Methods<br/> DOI: 10.1038/s41592-025-02814-z<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a><br/> <a href="#">Full text 2</a></p>                    | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-033   FORMAL VALIDATION<br/> <b>scCCVGBen for benchmarking of single-cell representation learning anchored on a centroid-coupled variational graph attention autoencoder across scRNA-seq and scATAC-seq</b><br/> 2026   Frontiers in Genetics<br/> DOI: 10.3389/fgene.2026.1822168<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-008   FORMAL VALIDATION<br/> <b>Towards Cross-Sample Alignment for Multi-Modal Representation Learning in Spatial Transcriptomics</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.03.02.709002<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                      | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-009   FORMAL VALIDATION<br/> <b>iPro-CSAF: identification of promoters based on convolutional spiking neural networks and spiking attention mechanism</b><br/> 2025   PeerJ Computer Science<br/> DOI: 10.7717/peerj-cs.2761<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                        | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-036   FORMAL VALIDATION<br/> <b>CART-GPT: A T Cell-Informed AI Linguistic Framework for Interpreting Neurotoxicity and Therapeutic Outcomes in CAR-T Therapy</b><br/> 2025<br/> DOI: 10.1101/2025.08.08.669387<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-050   FORMAL VALIDATION<br/> <b>Context-dependent utility and robustness of pretrained single-cell foundation model representations across analytical tasks</b><br/> 2026<br/> DOI: 10.64898/2026.06.18.733285<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-006   FORMAL VALIDATION<br/> <b>Artificial Intelligence agents for biological research: a survey</b><br/> 2026   Briefings in Bioinformatics<br/> DOI: 10.1093/bib/bbag075<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                     | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-032   FORMAL VALIDATION<br/> <b>OCOO-T : A SIMPLE AND SCALABLE VIRTUAL CELL MODEL FOR TRANSCRIPTIONAL PERTURBATION RESPONSE PREDICTION</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.06.08.731000<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-014   FORMAL VALIDATION<br/> <b>Residual-stream geometry of single-cell foundation models carries incremental gene-regulatory signal across tissues</b><br/> 2026   BMC Bioinformatics<br/> DOI: 10.1186/s12859-026-06538-5<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>         | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-037   FORMAL VALIDATION<br/> <b>Tahoe-x1: Scaling Perturbation-Trained Single-Cell Foundation Models to 3 Billion Parameters</b><br/> 2025<br/> DOI: 10.1101/2025.10.23.683759<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                      | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-016   FORMAL VALIDATION</p> <p><b>Investigation of cell development and tissue structure network based on natural Language processing of scRNA-seq data</b></p> <p>2025   Journal of Translational Medicine</p> <p>DOI: 10.1186/s12967-025-06263-2</p> <p><a href="#">Article page</a></p> <p><a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED</p> <p>Citation validity:</p> <p>Method executed:</p> <p>Primary use:</p> <p>Biological insight:</p> <p>Evidence sufficiency:</p> <p>Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-024   FORMAL VALIDATION</p> <p><b>Scouter predicts transcriptional responses to genetic perturbations with large language model embeddings</b></p> <p>2025   Nature Computational Science</p> <p>DOI: 10.1038/s43588-025-00912-8</p> <p><a href="#">Article page</a></p> <p><a href="#">Full text 1</a></p>                   | <p>Review status: IN PROGRESS / LOCKED</p> <p>Citation validity:</p> <p>Method executed:</p> <p>Primary use:</p> <p>Biological insight:</p> <p>Evidence sufficiency:</p> <p>Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-049   FORMAL VALIDATION</p> <p><b>Routine FFPE sections support clinically compatible single-nucleus transcriptomics across six human cancer types</b></p> <p>2026</p> <p>DOI: 10.64898/2026.07.23.740343</p> <p><a href="#">Article page</a></p> <p><a href="#">Full text 1</a></p>                                          | <p>Review status: IN PROGRESS / LOCKED</p> <p>Citation validity:</p> <p>Method executed:</p> <p>Primary use:</p> <p>Biological insight:</p> <p>Evidence sufficiency:</p> <p>Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-042   FORMAL VALIDATION<br/> <b>Foundation model reveals the shared organization of transcription and topologically associating domains</b><br/> 2026   Cell Systems<br/> DOI: 10.1016/j.cels.2026.101675<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                   | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-003   FORMAL VALIDATION<br/> <b>sciLaMA: A Single-Cell Representation Learning Framework to Leverage Prior Knowledge from Large Language Models</b><br/> 2025   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2025.01.28.635153<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-026   FORMAL VALIDATION<br/> <b>Cancer-associated fibroblasts enhance colorectal cancer lymphatic metastasis via CLEC11A/LGR5-mediated WNT pathway activation</b><br/> 2025   Journal of Clinical Investigation<br/> DOI: 10.1172/jci194243<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-019   FORMAL VALIDATION<br/> <b>Toward automated and explainable high-throughput perturbation analysis in single cells</b><br/> 2025   Patterns<br/> DOI: 10.1016/j.patter.2025.101228<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                   | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-046   FORMAL VALIDATION<br/> <b>AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes</b><br/> 2025   Briefings in Bioinformatics<br/> DOI: 10.1093/bib/bbaf426<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-027   FORMAL VALIDATION<br/> <b>Bioinformatics Copilot 2.0 for Transcriptomic Data Analysis</b><br/> 2024   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.1101/2024.08.15.607673<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                                  | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-043   FORMAL VALIDATION<br/> <b>DECIPHER for learning disentangled cellular embeddings in large-scale heterogeneous spatial omics data</b><br/> 2025   Nature Communications<br/> DOI: 10.1038/s41467-025-63140-8<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                                    | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-018   FORMAL VALIDATION<br/> <b>Cellfm-datasets: A Unified Data Infrastructure for Single-Cell and Spatial Transcriptomics Foundation Model Pretraining</b><br/> 2026   bioRxiv (Cold Spring Harbor Laboratory)<br/> DOI: 10.64898/2026.06.11.731508<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-030   FORMAL VALIDATION<br/> <b>A Foundation Model Should Not Flatten Pharmaceutical Evidence across Molecules, Sequences, Omics, Images, and Scientific Text</b><br/> 2024   Pharmacophore<br/> DOI: 10.51847/rbfe0o1wzp<br/> <a href="#">Article page</a><br/> <a href="#">Full text 1</a></p>                            | <p>Review status: IN PROGRESS / LOCKED<br/> Citation validity:<br/> Method executed:<br/> Primary use:<br/> Biological insight:<br/> Evidence sufficiency:<br/> Supporting quote + locator:</p> <p>Notes:</p> |

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| <p>HV-010   FORMAL VALIDATION</p> <p><b>evoCancerGPT: Generating Zero-Shot Single-Cell and Single-Sample Cancer Progression Through Transfer Learning</b></p> <p>2026   bioRxiv (Cold Spring Harbor Laboratory)</p> <p>DOI: 10.64898/2026.02.12.705621</p> <p><a href="#">Article page</a></p> <p><a href="#">Full text 1</a></p> | <p>Review status: IN PROGRESS / LOCKED</p> <p>Citation validity:</p> <p>Method executed:</p> <p>Primary use:</p> <p>Biological insight:</p> <p>Evidence sufficiency:</p> <p>Supporting quote + locator:</p> <p>Notes:</p> |
| <p>HV-044   FORMAL VALIDATION</p> <p><b>CellAgent: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis</b></p> <p>2024   bioRxiv (Cold Spring Harbor Laboratory)</p> <p>DOI: 10.1101/2024.05.13.593861</p> <p><a href="#">Article page</a></p> <p><a href="#">Full text 1</a></p>                    | <p>Review status: IN PROGRESS / LOCKED</p> <p>Citation validity:</p> <p>Method executed:</p> <p>Primary use:</p> <p>Biological insight:</p> <p>Evidence sufficiency:</p> <p>Supporting quote + locator:</p> <p>Notes:</p> |